

BOTHY · DEFENCE SUPPLY-CHAIN RESILIENCE

Which programmes break when gallium does?

Captured graph evidence, cited verification, and accountable follow-up for a European defence-prime supply-chain analyst.

European Defense Tech Hackathon · London · 1–4 October 2026 · Team 19 · public/synthetic prototype · github.com/thisyearnofear/bothy · live demo bothy.trustfall.xyz

A real deadline. No list of what it touches.

Beijing announced gallium export licensing on 3 July 2023, banned exports to the US in December 2024, then suspended the ban in November 2025. The suspension is scheduled to end **27 November 2026**. For a defence-prime analyst the recurring question is the same:

Which of our platforms depend on primary gallium — and who is checking inventory, qualified alternates and delivery timing before that date?

Trace

Modelled dependencies across multiple bill-of-material tiers.

Inspect

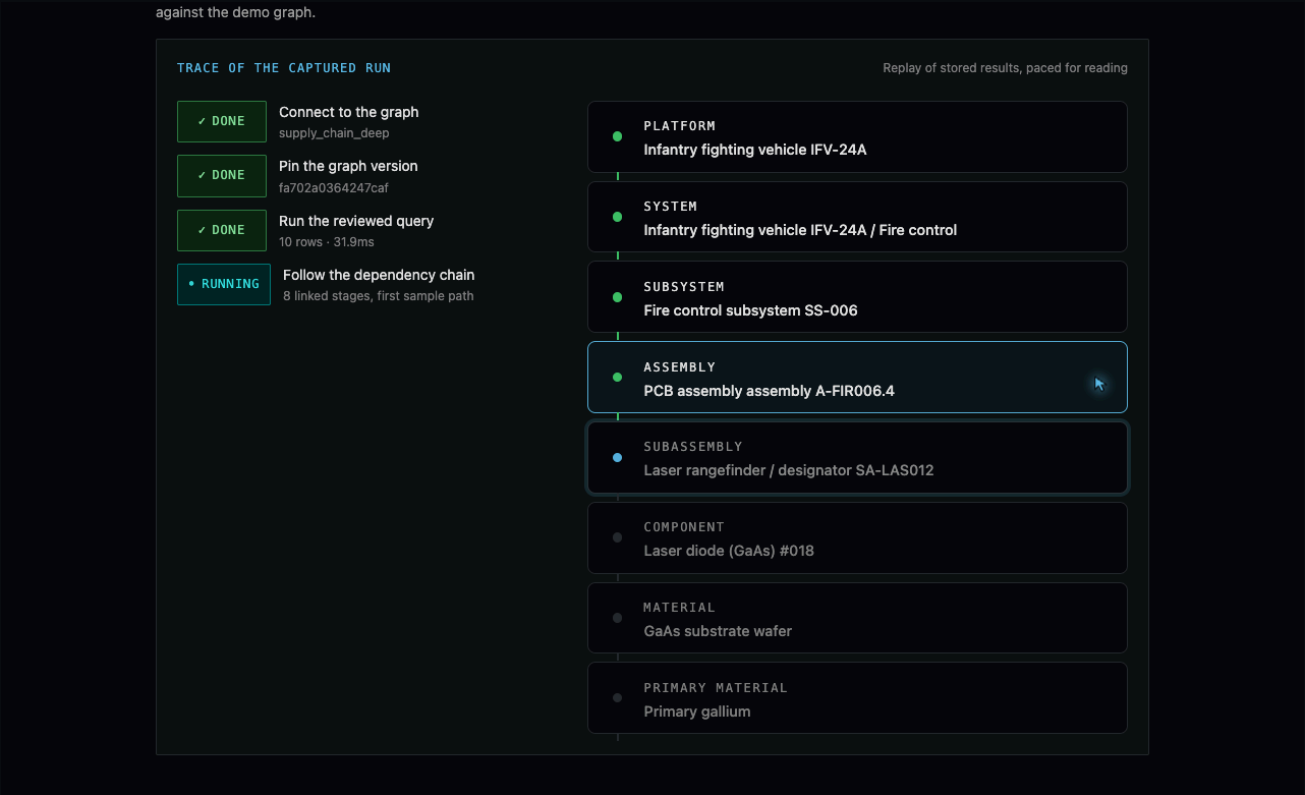
Pinned graph revision, cited rows and coverage gaps stay with the claim.

Own

A named reviewer approves; a named owner reports the finding.

USGS 2026: China accounted for ~99% of worldwide primary low-purity gallium production. Gallium is on the European Commission's 2023 critical raw materials list. Neither fact verifies any synthetic platform's exposure.

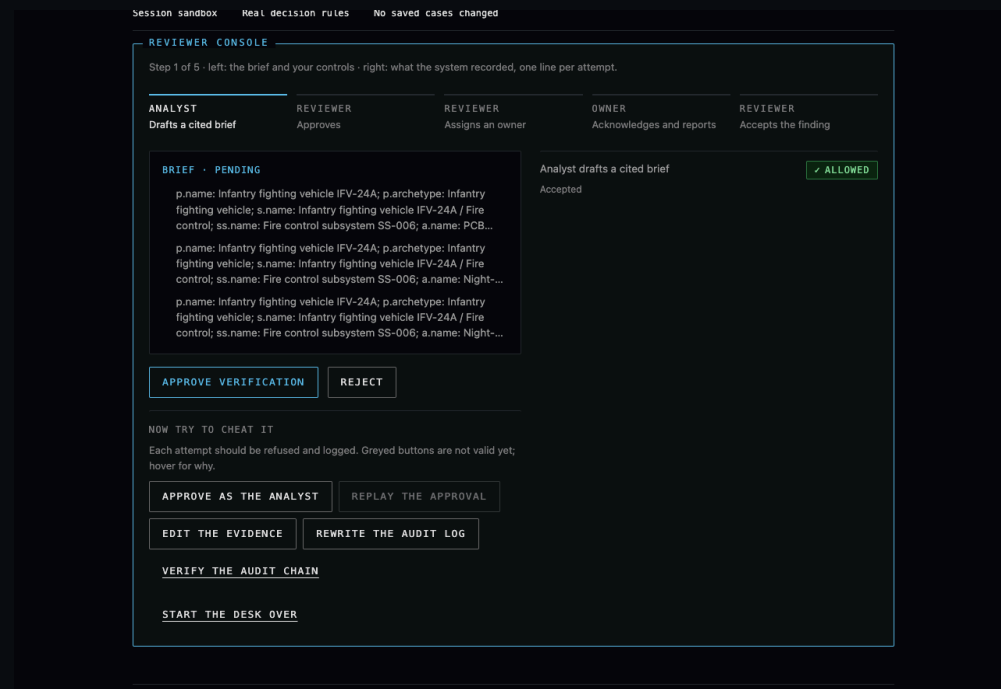
From a question to a pinned, inspectable chain



A reviewed catalogue query against the versioned TuringDB supply-chain graph: connect → pin revision fa702a0364247caf → run → follow the typed chain Platform → System → Subsystem → Assembly → Subassembly → Component → Material → Primary gallium. 10 sampled rows in ~32ms; the bounded exposure question separately observed 34 platform names (limit 50). Synthetic names; sampled rows are not a total exposure count.

Every step has a role. Every attempt is recorded.

- **Reviewed reads only.** An exact query allowlist; arbitrary Cypher and writes are refused before reaching the graph. Fresh isolated client per read; HEAD resolves to a concrete pinned revision with query hash.
- **Deterministic cited briefs.** Each claim links to its stored evidence row; the brief is bound to the hash of its evidence.
- **Real roles, fail-closed.** Analyst drafts; reviewer approves and assigns; named owner acknowledges and reports. OIDC-verified — no role, no decision.
- **Hash-chained audit.** Each entry commits to the one before it; a rewrite surfaces at the exact entry.



The interactive reviewer desk lets a jury try to cheat it: self-approve, replay the approval, acknowledge as the wrong person, edit the evidence, rewrite the audit log — each attempt is refused or detected against the real decision rules on a throwaway store.

WHAT WAS BUILT AT THE EVENT

Built inside the window

SCOPE	WORK
Pre-event baseline through 15 Sep (678ab41)	Earlier winter-road/flood demos only: deterministic scoring, review gate, Postgres ledger. No defence workspace, no graph integration.
In the build window 2–4 Oct · 35 commits · +11,700 lines	One-command venue stack; TuringDB sidecar with 9-scenario reviewed catalogue, pinned reads, replay/diff and witness packs; browser SSO bridge (PKCE) with OIDC-verified review → assign → acknowledge → outcome under transactional audit; scoped work views, saved cases and linked revisions; guided two-minute demo player; date-scrubbed real-event stories (gallium deadline, Red Sea diversion); interactive reviewer desk; six-attack stress-test lab; run trace; VPS deployment with hosted invite-gated SSO.
Evidence	150 TypeScript tests + 6 Python graph-boundary tests pass; typecheck, lint and production build clean. Live rehearsal on the submission revision ran the real sidecar, agent and UI against the graph at revision fa702a0364247caf.

Boundary commits and test output are in the repository history and docs/edth-submission-pack.md.

DEPLOYMENT, BOUNDARIES, NEXT PROOF

A focused prototype; a measurable pilot next

Next.js interface · TypeScript agent · SQLite case records ·
Python/TuringDB sidecar.

- Synthetic and public evidence only; no real buyer identity provider validated.
- Not certified offline, tenant-isolated, or production-ready; recorded outcomes are not proven operational impact.
- Dependencies in the demo graph do not establish programme stoppage.

Next proof:

- One buyer-approved reference dataset and impact question.
- Analyst-checked dependency accuracy and coverage.
- Measured time to a review-ready brief vs the current process.

Ask: a defence-prime design partner to validate the recurring decision, plus an identity and data-handling review.

[Repository](#) · [Live demo](#) · recorded-demo link added at submission